



FINAL REPORT - Study # 174-001

Prepared by: Genemarkers

Differential Gene Expression Analysis of Mouse Lung Tissues

Project Leader: Kelly Boddington

Study Sponsor: Canurta

Approved by: _____

A handwritten signature in black ink, appearing to be "ajl", written over a horizontal line.

Date: _____

9/28/2023

Study Information

Customer Name	Kelly Boddington
Customer Company/Institution	Canurta
Customer Email	kelly@canurta.com
Report Submission Date	27SEP2023
Study #	174-001
Tissue Type	Mouse Lung Tissue
Reference Mouse Genome	mm9
Library Preparation Kit	Illumina Stranded mRNA Prep
Type of Sequencer	NextSeq-550Dx
Read Length	75 (single end)
Sequencing Depth	10 million per sample
Number of Samples	12

Additional files (provided as 174-001 BSSH+DAVID 22SEP2023DA.zip file) for each comparison has the following:

- 1.** Genes showing differential expression in each comparison
- 2.** List of genes for all significantly enriched biological processes in each comparison
- 3.** List of genes for all significantly enriched KEGG pathways in in each comparison

Introduction to RNA-Seq

Ribonucleic acid sequencing (RNA-Seq) uses next-generation sequencing (NGS) to enable identification and quantification of messenger RNA (mRNA) molecules present in a biological sample at a given point in time. The set of RNA molecules transcribed in a certain condition and time reflects the existing condition of a cell and can reveal important molecular mechanisms governing a disease or a particular phenotype. mRNA-Seq facilitates the ability to look at changes in global gene expression patterns over time and differences in gene expression in different treatment groups.

RNA-Seq is a high-throughput technology that offers several advantages, including higher specificity, sensitivity, and broader dynamic range over hybridization-based techniques, such as microarrays. Additionally, differential gene expression analysis based on RNA-Seq does not require *a priori* knowledge of the nucleotide sequence of the target organism, such as qPCR. In practice, millions of small stretches, called 'reads' are sequenced from different positions of the input RNAs which are then quality checked and mapped back to the reference genome. The number of reads aligned to each gene provides a measure of its level of expression. The read count data is then normalized using statistical models to identify differentially expressed genes (DEGs). The data is then assessed within a biological framework.

Project Summary

The study was performed to understand the effects of the Sponsor's lead molecule on acute respiratory distress syndrome (ARDS) in mouse lung tissues. Lipopolysaccharide (LPS)-induced lung injury mouse model tissues were treated, collected by the Sponsor, and shipped on dry ice to Genemarkers for differential gene expression analysis. The study design included two test material (TM) groups and a control group, four replicates per group (N=4; listed in Table 1). Sequencing was performed using Illumina's NextSeq 550Dx instrument. Differential gene expression analysis and additional biological pathway analyses were performed using the bioinformatics software programs listed in Table 2.

Table 1: Treatment Group and Genemarkers Sample ID Assignment

Group #	Group ID	Treatment	Sample IDs
1	Veh	Negative Control	H23-142 to H23-145
2	TM A	Cannaflavin A	H23-146 to H23-149
3	TM B	Cannaflavin B	H23-150 to H23-153

Table 2: Bioinformatics Software Programs Used for Data Analysis

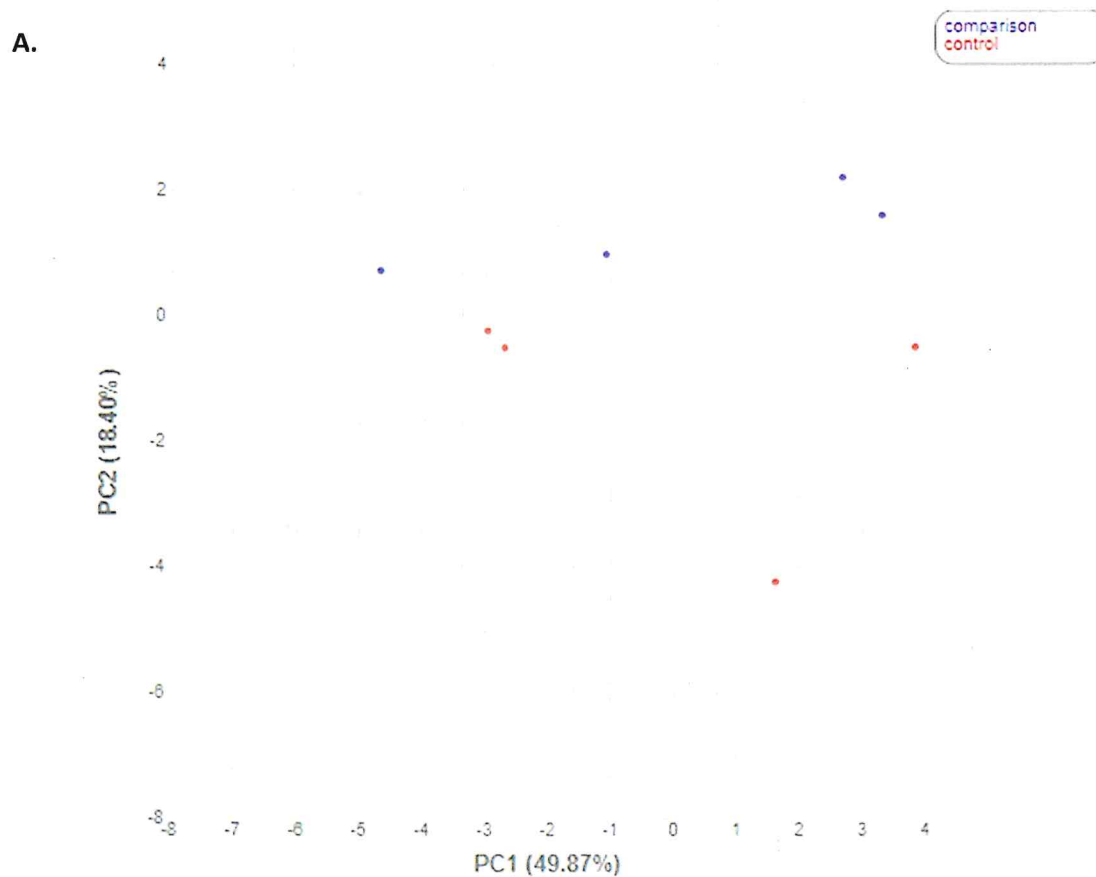
Type of analysis	Software/app	Version
FASTQ conversion	BaseSpace BCL Convert	2.2.0
5' T-base trimming	BaseSpace FASTQ Toolkit	2.2.5
Quality control and Mapping	BaseSpace RNA-Seq Alignment	2.0.2
Differential gene expression analysis	BaseSpace RNA-Seq Differential Expression	1.0.1
Heatmaps	Heatmapper	NA
Pathway analysis	DAVID	2021q4

Results

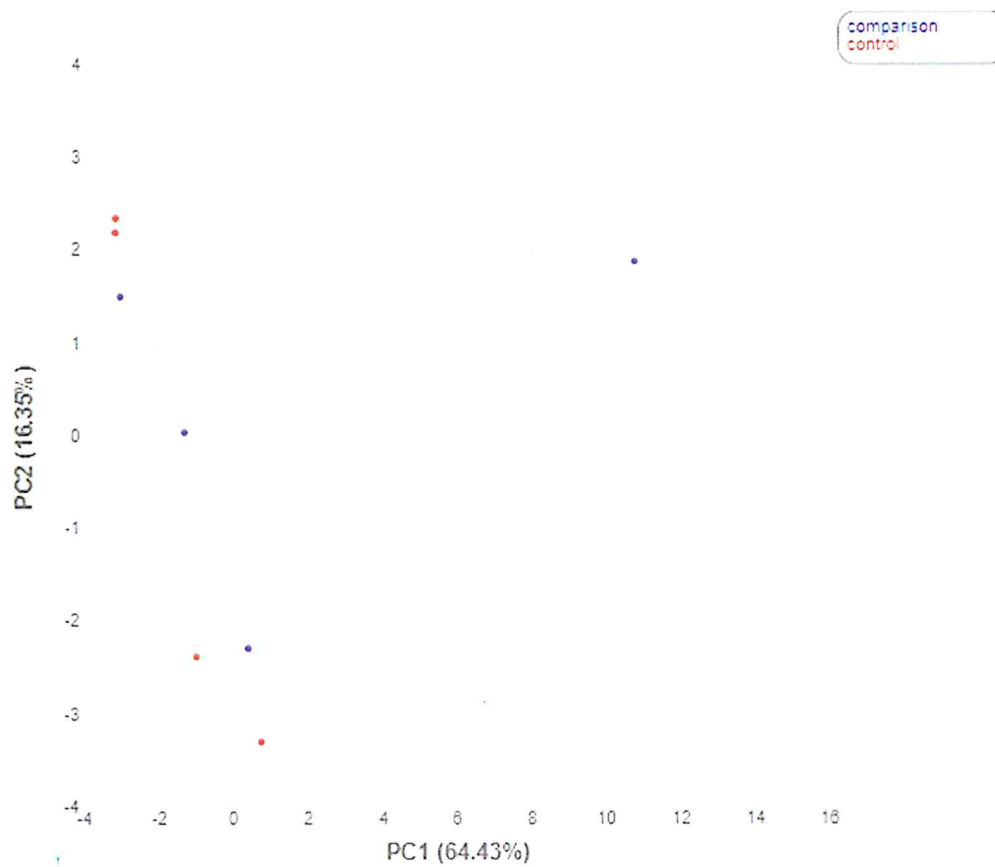
Principal Component Analysis

Principal Component Analysis (PCA) plots are used to visualize the data and identify patterns between test groups or variables. Samples with similar gene expression profiles cluster together while divergent samples separate. Ideally, all replicates of a group should cluster together.

Figure 1. PCA plots for **A.** TMA vs Veh and **B.** TMB vs Veh



B.



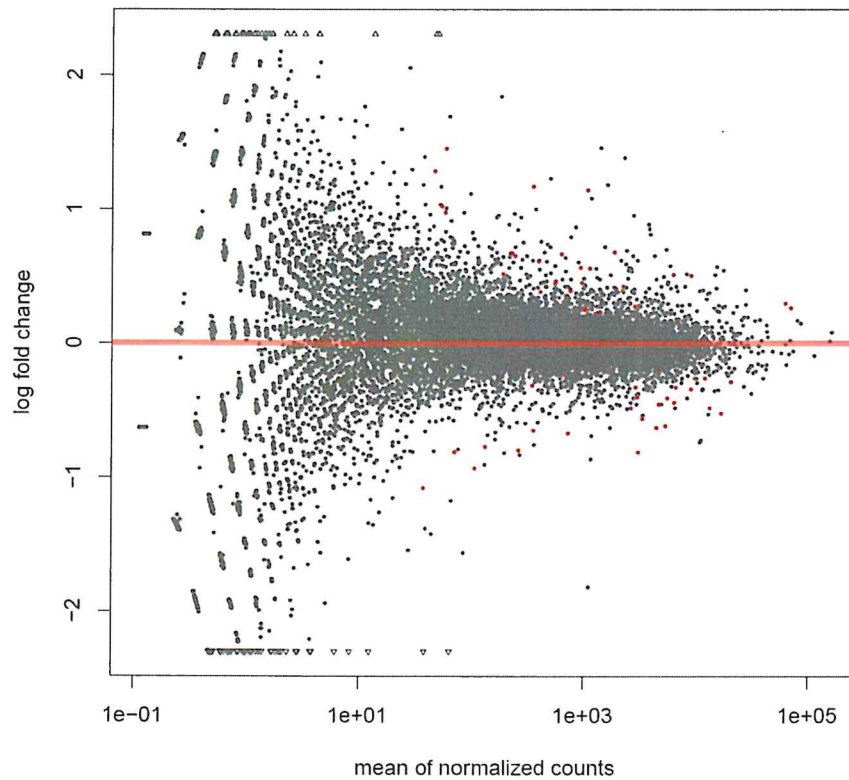
Differential Gene Expression Analysis

Table 3: Number of differentially expressed genes in different comparisons using a p-value of 0.05, and log2fold change cutoff of 0.6.

Comparison roup	Total number of DEGs	Up-regulated	Down-regulated
TMA vs Veh	165	117	48
TMB vs Veh	175	147	28

Figure 2. MA plots display genes that are differentially expressed (p.adj. < 0.05) in **A.** TMA vs Veh and **B.** TMB vs Veh Red dots represent differentially expressed genes while grey dots indicate individual genes that are not differentially expressed

A.



B.

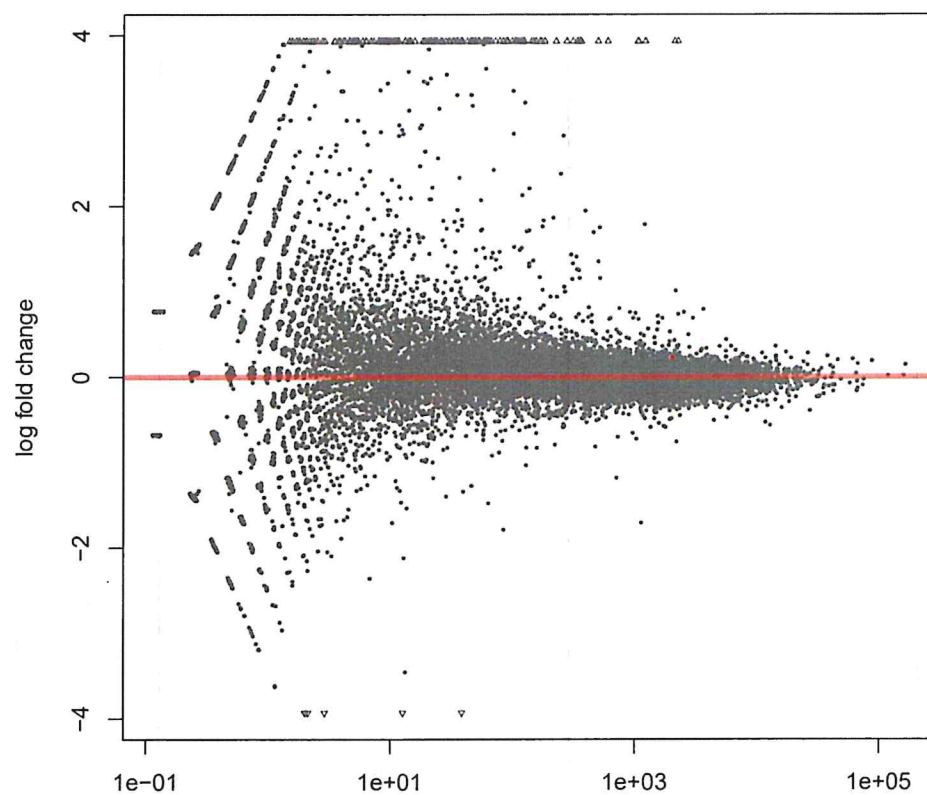
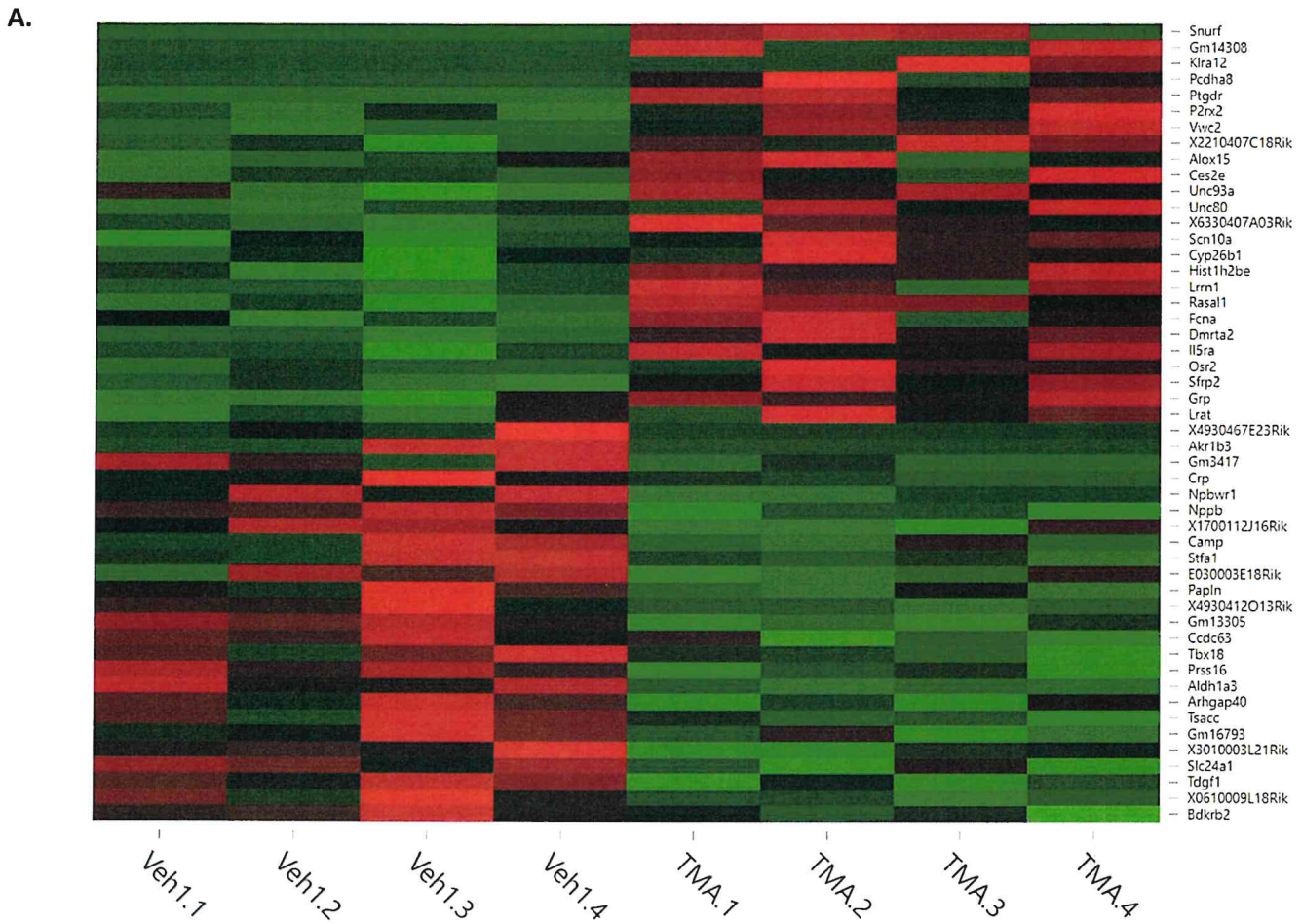


Figure 3. Heat maps showing top 50 significantly (p val.< 0.05) up-and down-regulated DEGs for **A.** TMA vs Veh and **B.** TMB vs Veh.



B.

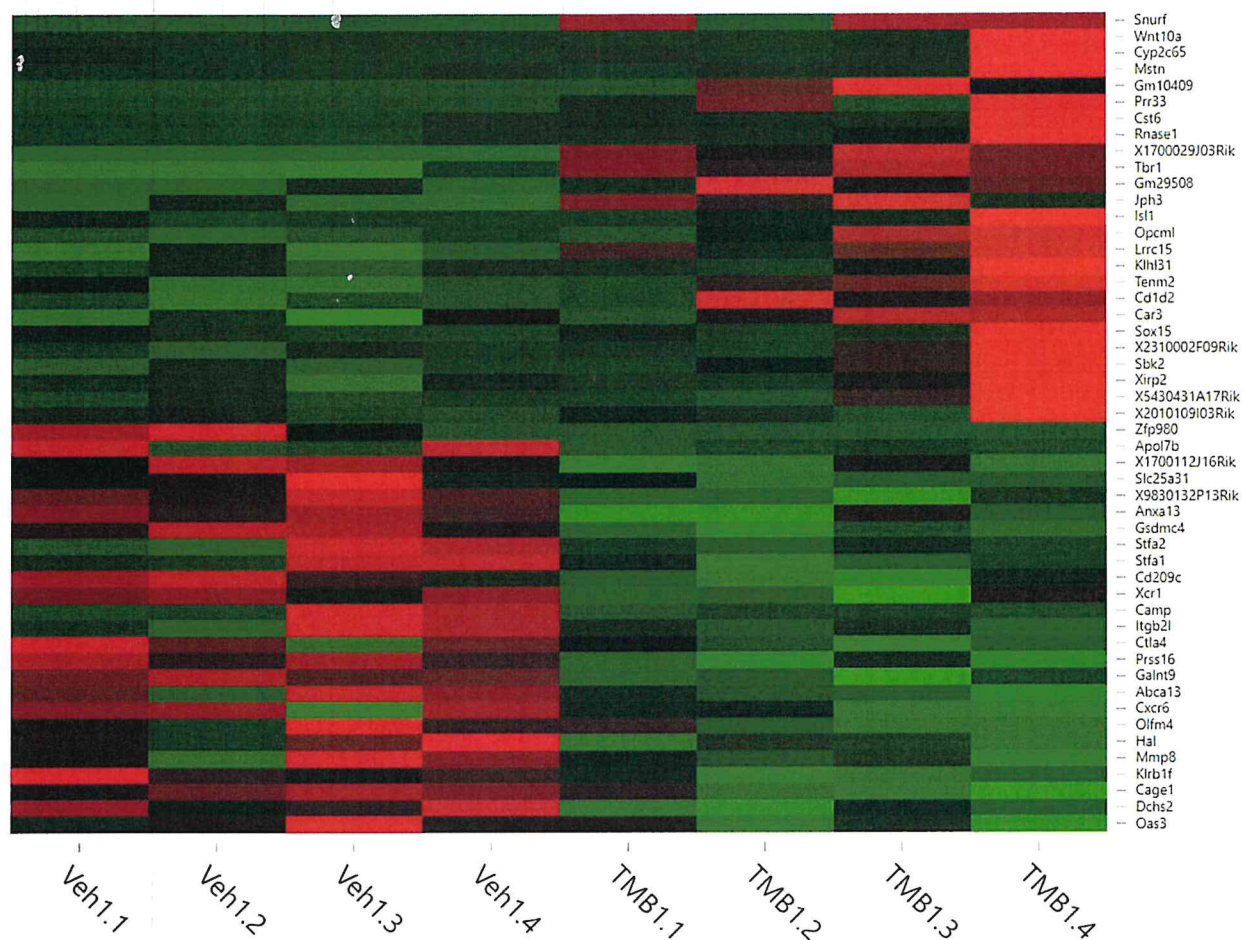


Table 4A. Significantly (p-val. < 0.05) enriched Gene Ontology Terms in Biological Process Category for DEGs in Treatment A when Compared with Vehicle Control.

Direction	P-value	# Genes	Biological Process	Genes
TMA vs Veh				
Up-regulated	1.8E-04	12	GO:0007155~cell adhesion	NLGN3, AOC3, SPON2, MFAP4, COL15A1, COL14A1, LAMA1, CLDN8, NRCAM, PCDHA8, EPHB1, PCDHAC2
	2.3E-04	6	GO:0001503~ossification	IGSF10, FOXC2, CLEC11A, ALOX15, STC1, GFRA4
	2.8E-03	5	GO:0007169~transmembrane receptor protein tyrosine kinase signaling pathway	PTPRT, NTRK2, FGFR4, EPHB1, GFRA4
	5.6E-03	4	GO:0051965~positive regulation of synapse assembly	NLGN3, NTRK2, LRRN1, EPHB1
	8.7E-03	3	GO:0048266~behavioral response to pain	SCN10A, P2RX2, OSM
	1.3E-02	3	GO:0032635~interleukin-6 production	SPON2, BANK1, PRG4
	1.4E-02	9	GO:0006468~protein phosphorylation	NTRK2, SRPK3, CDK19, PHKG1, LAMA1, CDK1, FGFR4, EPHB1, AURKB
	1.4E-02	4	GO:0007623~circadian rhythm	NTRK2, SCN10A, NOS2, IGF1
	1.6E-02	3	GO:0060349~bone morphogenesis	OSR2, CYP26B1, SFRP2
	2.0E-02	3	GO:0007616~long-term memory	NTRK2, EGR1, SHANK1
	2.1E-02	3	GO:0010811~positive regulation of cell-substrate adhesion	ABI3BP, ALOX15, VWC2
	2.4E-02	2	GO:0048866~stem cell fate specification	DMRTA2, SFRP2
	2.4E-02	2	GO:0043152~induction of bacterial agglutination	SPON2, DMBT1
	2.5E-02	3	GO:0001974~blood vessel remodeling	FOXC2, NOS2, IGF1
	3.0E-02	5	GO:0070374~positive regulation of ERK1 and ERK2 cascade	CCL11, ALOX15, CCL4, IGF1, FGFR4
	3.2E-02	3	GO:0030199~collagen fibril organization	FOXC2, SFRP2, COL14A1
	3.2E-02	8	GO:0016310~phosphorylation	NTRK2, SRPK3, CDK19, PHKG1, CDK1, FGFR4, EPHB1, AURKB
	3.7E-02	8	GO:0008284~positive regulation of cell proliferation	GLP1R, OSR2, NTRK2, SFRP2, CLEC11A, OSM, IGF1, FGFR4

Direction	P-value	# Genes	Biological Process	Genes
	4.1E-02	3	GO:0035176~social behavior	NLGN3, GRP, SHANK1
	4.3E-02	6	GO:0010629~negative regulation of gene expression	SFRP2, NOS2, APLNR, CDK1, IGF1, FGFR4
Down-regulated	5.5E-04	3	GO:0042310~vasoconstriction	CRP, BDKRB2, SMAD6
	6.3E-03	2	GO:0070384~Harderian gland development	ALDH1A3, SOX9
	7.4E-03	3	GO:0071347~cellular response to interleukin-1	AKAP12, SOX9, CAMP
	1.1E-02	2	GO:0072189~ureter development	SOX9, TBX18
	1.8E-02	5	GO:0008284~positive regulation of cell proliferation	CNTFR, CDCA7L, SOX9, TDGF1, CAMP
	1.9E-02	2	GO:0003170~heart valve development	SOX9, SMAD6
	1.9E-02	2	GO:0060013~righting reflex	ALDH1A3, CDH23
	2.3E-02	2	GO:0035810~positive regulation of urine volume	NPPB, NPR3
	2.9E-02	6	GO:0030154~cell differentiation	CCDC63, RORC, SOX9, SMAD6, TDGF1, TBX18
	3.4E-02	2	GO:0010463~mesenchymal cell proliferation	SOX9, TBX18
	3.6E-02	2	GO:0009948~anterior/posterior axis specification	TDGF1, TBX18
	4.3E-02	2	GO:0090103~cochlea morphogenesis	SOX9, TBX18
	4.3E-02	2	GO:0003180~aortic valve morphogenesis	SOX9, SMAD6
	4.6E-02	6	GO:0006355~regulation of transcription, DNA-templated	CDCA7L, RORC, SOX9, ZFP641, SMAD6, TBX18
	4.8E-02	2	GO:0030279~negative regulation of ossification	SOX9, SMAD6

Table 4B. Significantly (p-val. < 0.05) enriched Gene Ontology Terms in Biological Process Category for DEGs in Treatment B when Compared with Vehicle Control.

Direction	P-value	# Genes	Biological Process	Genes
TMB vs Veh				
Up-regulated	1.8E-07	7	GO:0045214~sarcomere organization	CSRP3, XIRP1, TCAP, MYPN, MYOM2, TTN, LMOD2
	2.2E-05	6	GO:0006936~muscle contraction	MYOM1, TPM2, LMOD3, MYOM2, TTN, LMOD2

Direction	P-value	# Genes	Biological Process	Genes
	2.0E-04	3	GO:0035995~detection of muscle stretch	CSRP3, TCAP, TTN
	5.0E-04	3	GO:0030240~skeletal muscle thin filament assembly	TCAP, LMOD3, TTN
	8.5E-04	4	GO:0055013~cardiac muscle cell development	MYO18B, XIRP1, TCAP, TTN
	9.1E-04	12	GO:0010628~positive regulation of gene expression	TFAP2A, OSR2, NTRK2, MYOM1, WNT10A, CYP26B1, WNT7B, MST1, FGFR4, MT3, SOX10, TTN
	2.5E-03	3	GO:0055003~cardiac myofibril assembly	CSRP3, TCAP, TTN
	2.8E-03	6	GO:0007015~actin filament organization	PDLIM3, TPM2, XIRP1, XIRP2, LMOD3, LMOD2
	2.9E-03	3	GO:0003300~cardiac muscle hypertrophy	CSRP3, TCAP, TTN
	3.8E-03	3	GO:0055008~cardiac muscle tissue morphogenesis	TCAP, XIRP2, TTN
	4.9E-03	3	GO:0010880~regulation of release of sequestered calcium ion into cytosol by sarcoplasmic reticulum	HRC, GSTM7, TRDN
	5.9E-03	5	GO:0045471~response to ethanol	CAR3, MSTN, ADIPOQ, SLC2A4, CYP2E1
	8.2E-03	7	GO:0009617~response to bacterium	CAR3, FABP4, ADIPOQ, LRAT, CYP2E1, RNASE1, TRDN
	1.3E-02	14	GO:0030154~cell differentiation	OSR2, NTRK2, TENM4, PAQR5, WNT7B, SPINK5, ISL1, SOX10, NPAS4, GAP43, CSRP3, SOX15, RXRG, TRIM54
	1.4E-02	3	GO:0043403~skeletal muscle tissue regeneration	MSTN, SGCA, SOX15
	1.5E-02	3	GO:0048738~cardiac muscle tissue development	CSRP3, SGCG, TTN
	1.6E-02	6	GO:0030182~neuron differentiation	NTRK2, WNT10A, WNT7B, TBR1, RXRG, ISL1
	1.7E-02	2	GO:0048769~sarcomerogenesis	TCAP, TTN
	2.3E-02	3	GO:0042476~odontogenesis	OSR2, WNT10A, PHEX
	2.3E-02	3	GO:0060349~bone morphogenesis	TFAP2A, OSR2, CYP26B1
Down-regulated	2.6E-04	6	GO:0006955~immune response	XCR1, CD209C, CTLA4, CXCR6, OLFM4, CAMP
	1.7E-03	4	GO:0098609~cell-cell adhesion	ITGB2L, DCHS2, STFA1, STFA2
	1.6E-02	2	GO:0050777~negative regulation of immune response	CTLA4, OLFM4

Direction	P-value	# Genes	Biological Process	Genes
	3.9E-02	2	GO:0018149~peptide cross-linking	STFA1, STFA2
	4.5E-02	2	GO:0045861~negative regulation of proteolysis	STFA1, STFA2

Table 5A. Significantly (p-val. < 0.05) enriched KEGG Pathways for DEGs in Treatment A when Compared with Vehicle Control.

Direction	P-value	# Genes	KEGG Pathway	Genes
TMA vs Veh				
Up-regulated	0.031	5	mmu04014:Ras signaling pathway	NTRK2, GNG7, RASAL1, IGF1, FGFR4
	0.032	4	mmu04371:Apelin signaling pathway	EGR1, NOS2, GNG7, APLNR
	0.035	6	mmu04151:PI3K-Akt signaling pathway	NTRK2, LAMA1, GNG7, OSM, IGF1, FGFR4
	0.039	5	mmu04020:Calcium signaling pathway	NTRK2, PHKG1, NOS2, P2RX2, FGFR4
Down-regulated	None			

Table 5B. Significantly (p-val. < 0.05) enriched KEGG Pathways for DEGs in Treatment B when Compared with Vehicle Control.

Direction	P-value	# Genes	KEGG Pathway	Genes
TMB vs Veh				
Up-regulated	5.7E-05	9	mmu04020:Calcium signaling pathway	NTRK2, SLN, PHKG1, CAMK2A, MST1, HRC, FGFR4, TRDN, MYLK4
	1.5E-03	5	mmu05410:Hypertrophic cardiomyopathy	SGCA, TPM2, SGCG, ACE3, TTN
	1.2E-02	4	mmu04260:Cardiac muscle contraction	TPM2, HRC, COX7A1, TRDN
	1.4E-02	4	mmu05414:Dilated cardiomyopathy	SGCA, TPM2, SGCG, TTN
	2.8E-02	3	mmu04930:Type II diabetes mellitus	KCNJ11, ADIPOQ, SLC2A4
	3.0E-02	3	mmu00591:Linoleic acid metabolism	PLA2G4F, CYP2C65, CYP2E1
	4.0E-02	4	mmu04550:Signaling pathways regulating pluripotency of stem cells	WNT10A, WNT7B, FGFR4, ISL1

Direction	P-value	# Genes	KEGG Pathway	Genes
Down-regulated	0.018597638	3	mmu05152:Tuberculosis	ITGB2L, CD209C, CAMP
	0.024403139	3	mmu04613:Neutrophil extracellular trap formation	ITGB2L, SLC25A31, CAMP

Experimental Procedures Summary

Sample receipt: Twelve frozen mouse lung tissue samples were received at Genemarkers on 23 August, 2023. The samples were stored at -80°C until RNA extraction.

RNA extraction, quantification and qualification: Total RNA extractions were performed using a Maxwell RSC SimplyRNA kit (Promega, WI) with the addition of a Proteinase K incubation. RNA was eluted in a final volume of 40 µl. RNA concentration and purity were determined using a Nanodrop 2000 spectrophotometer and a Qubit flex fluorometer using the BR RNA kit (ThermoFisher Scientific, MA, USA). The integrity of RNA was analyzed using a Qubit™ RNA IQ Assay Kit. RNA samples showing absorbance ratios of 260:280 nm and 260:230 nm above 2, and IQ# above 7 are criteria used for sequencing (Table 6). All samples met the required criteria metrics.

Table 6: RNA Quantification and Qualification Metrics.

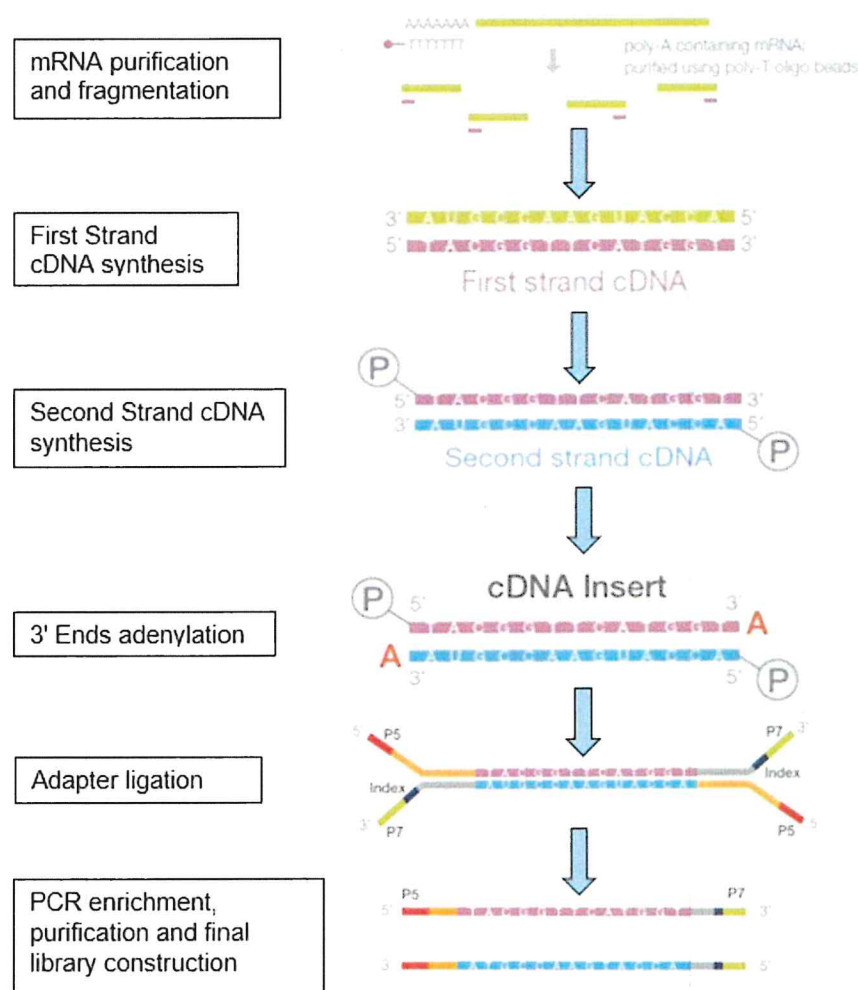
Sample ID	Group ID	Nanodrop ng/µL	Qubit ng/ µL	260/280	260/230	RNA IQ
H23-142	Veh	468.4	344	2.08	2.16	8.7
H23-143		488.4	354	2.08	2.17	8.4
H23-144		356.3	228	2.1	0.89	8.2
H23-145		471	330	2.08	1.95	8.3
H23-146	TM A	980.1	740	2.09	2.23	7.6
H23-147		512.9	360	2.1	1.55	8.2
H23-148		524.2	384	2.09	2.22	8.4
H23-149		777.2	596	2.08	2.23	7.8
H23-150	TM B	418.3	318	2.09	2.15	8.1
H23-151		464	340	2.09	2.13	8.1
H23-152		563.9	430	2.08	2.08	7.8
H23-153		739.3	564	2.08	2.16	7.7

Library preparation and sequencing

Poly(A) RNA sequencing libraries were prepared following Illumina's Stranded-mRNA library preparation (Illumina, CA) protocol, with the exception of additional drying times (5min) for ethanol in the cleanup steps. Poly(A) tail-containing mRNAs were purified using oligo-(dT) magnetic beads and fragmented using fragmentation master mix in elevated temperature (94°C). The fragments were then used to synthesize first-strand and second-strand cDNA using random primers and reverse transcriptase. An 'A'

base was then ligated to the 3' ends of the double-stranded cDNA molecules followed by the addition of the Illumina sequencing adaptors. The cDNA fragments were then enriched using PCR and purified to create the final cDNA library. Fig. 4 summarizes the primary steps involved in the library preparation protocol. Quality control (QC) analysis of the sequencing library was performed using DNA 1000 and High Sensitivity DNA Chips on an Agilent Technologies 2100 Bioanalyzer instrument. Single-ended sequencing was performed on Illumina's Next-seq 550 Dx sequencing system using 1000 ng of input RNA.

Figure 4. Illumina library construction workflow.



Differential gene expression analysis

Differential gene expression analysis was conducted using several apps in Illumina's BaseSpace sequence hub. The BCL files from the sequencer were converted into FASTQ format using the BCL convert app. The adapter trimming option setting was utilized while setting up the run in Local Run Manager in the sequencer. The T' overhangs from FASTQ files were trimmed using FASTQ Toolkit v 2.2.5. The trimmed reads were then aligned to the mm9 mouse reference genome using the RNA-Seq alignment v 2.0.2

application. The RNA-Seq Differential Expression application v 1.0.1 was then used to run the DESeq2 algorithm on RNA quantification files produced by the RNA-Seq alignment app. This identifies genes and transcripts that are differentially expressed between the treatment group and control group. The differentially expressed genes (DEGs) were identified using a p-value cutoff of < 0.05 and linear fold change cutoff of >1.5 or < -1.5 ($\text{Log}_2\text{FC} > 0.6$ or < -0.6).

Functional enrichment and pathway analysis

Tertiary analysis of RNA-Seq data was performed using Database for Annotation, Visualization and Integrated Discovery (DAVID) (<https://david.ncifcrf.gov>). The DEGs identified between treatment vs control group were selected for gene ontology (GO) enrichment and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analyses. Following a separate enrichment analysis strategy, up- and down-regulated genes were uploaded separately to DAVID for pathway enrichment. Of the three major functional classes of the GO database, pathways related only to biological process (BP) are included in this report. Biological Processes and KEGG pathways with p-value ≤ 0.05 were considered significantly enriched.

Run QC Metrics

During the first few steps of the sequencing run, the clusters of cDNA fragments in the sequencing libraries are amplified in a process called cluster generation, which results in millions of copies of single-stranded DNA. The information coded in these single stranded DNAs is then read by the sequencer through a process called sequencing-by-synthesis (SBS). Cluster density is a critically important metric that influences sequencing performance in terms of data quality and total data output. Table 7 provides details on percentage of clusters passing filter, quality scores and percentages of cluster passing filters that are indexed or unindexed. The sequencing run metrics for this study met all of the required criteria suggested by Illumina.

Table 7. Sequencing Run Metrics.

Run Metrics	Value
Cluster density	142±4
% PF	92.43
Q30%	95.76
Yield (Gbp)	30.52
% Reads identified	98.22
% Reads undetermined	1.77

Cluster density: Number of clonal clusters generated by the binding of the template DNA to the surface of a flow cell

% PF: Percentage of clusters passing filter, an indicator of signal purity from each cluster

Q30%: percentage of bases with a Phred quality score of 30 or higher (Higher the Q30% better is the read data)

Yield: data output from a NextSeq 550 75-cycle high output kit

% Reads identified: Percentage of clusters passing filter that are assigned to an index

% Reads undetermined: Percentage of clusters passing filter that are not assigned to an index

Reference mapping and assembly

The trimmed reads obtained from FASTQ Toolkit were used to map against the mm9 mouse reference genome. The sample IDs, total number of input reads, and percent of mapped and unmapped reads along with quality scores are provided in Table 8.

Table 8. Mapping Data Statistics.

Sample ID	Input Reads ^a (M)	Mapped Reads ^b (M)	Unmapped Reads ^c (%)	Q30 (%)
H23-142	28.4	98.3%	1.7%	96.5%
H23-143	27.7	98.2%	1.8%	96.4%
H23-144	29.9	98.3%	1.7%	96.4%
H23-145	30.4	98.5%	1.5%	96.5%
H23-146	30	98.0%	2.0%	96.4%
H23-147	26.1	98.4%	1.6%	96.3%
H23-148	26.2	98.3%	1.7%	96.2%
H23-149	26.4	98.3%	1.7%	96.2%
H23-150	27.8	98.1%	1.9%	96.3%
H23-151	25.3	98.3%	1.7%	95.7%
H23-152	29.6	98.4%	1.6%	96.3%
H23-153	28.5	98.5%	1.5%	96.3%

^aInput reads: Total number of trimmed reads

^bMapped reads %: Percentage of input reads mapped to the reference genome

^cUnmapped reads %: Percentage of reads not mapped to the reference genome

^dQ30 (%): percentage of bases with a Phred quality score of 30 or higher

References:

1. Chu, Y. and Corey, D.R., 2012. RNA sequencing: platform selection, experimental design, and data interpretation. *Nucleic acid therapeutics*, 22(4), pp.271-274.
2. Finotello, F. and Di Camillo, B., 2015. Measuring differential gene expression with RNA-seq: challenges and strategies for data analysis. *Briefings in functional genomics*, 14(2), pp.130-142.
3. Whitley, S.K., Horne, W.T. and Kolls, J.K., 2016. Research techniques made simple: methodology and clinical applications of RNA sequencing. *Journal of Investigative Dermatology*, 136(8), pp.e77-e82.
4. Luo, W., Friedman, M.S., Shedden, K., Hankenson, K.D. and Woolf, P.J., 2009. GAGE: generally applicable gene set enrichment for pathway analysis. *BMC bioinformatics*, 10(1), pp.1-17.

